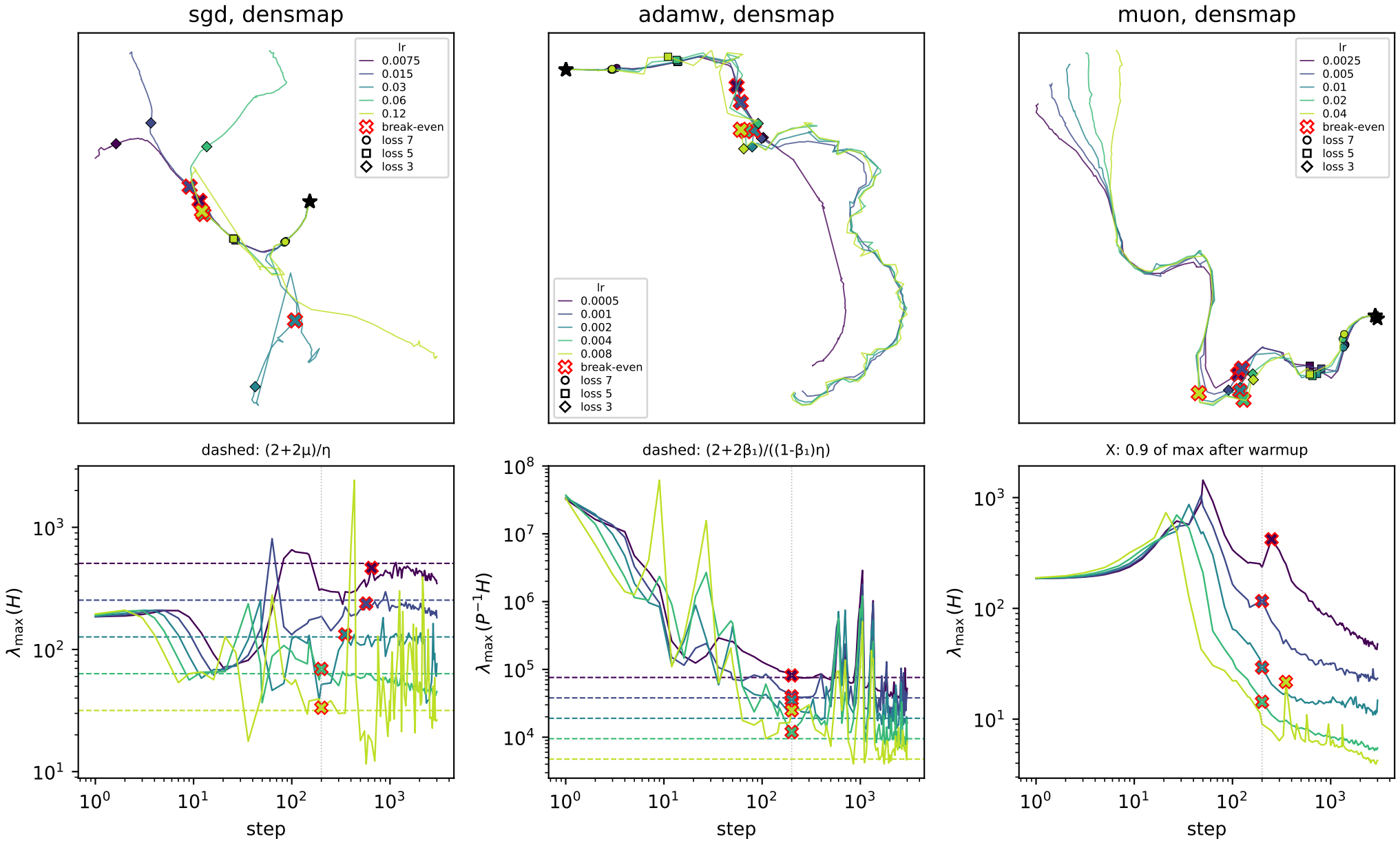

sgd, densmap
adamw, densmap
muon, densmap
lr
0.0075
0.015
0.03
0.06
0.12
break-even
loss 7
loss 5
loss 3
lr
0.0005
0.001
0.002
0.004
0.008
break-even
loss 7
loss 5
loss 3
lr
0.0025
0.005
0.01
0.02
0.04
break-even
loss 7
loss 5
loss 3
dashed: (2+2μ)/η
dashed: (2+2β₁)/((1-β₁)η)
X: 0.9 of max after warmup
$\lambda_{max}(H)$
$\lambda_{max}(P^{-1}H)$
$\lambda_{max}(H)$
step
step
step